

RKSI/ICN
INCHEON INTL

20 JAN 23


JEPPESSEN
SEOUL/INCHEON, KOREA
20-3
Eff 25 Jan 1600Z
SID

RADIO COMMUNICATION FAILURE PROCEDURE

In VMCs:

- 1) Squawk 7600.
- 2) Continue to fly in VMC.
- 3) Land at nearest suitable aerodrome.

Procedure for VFR flights

VFR flight which has encountered radio communication failure shall:

- 1) Squawk 7600, and
- 2) If able to see the light gun signal from control tower, follow that instruction.
- 3) If unable to see the light gun signal from control tower, hold over downwind for RWY 16R/34L, 15L/33R until ETA or for 10 minutes, whichever is longer; then
- 4) Land on RWY 16R/34L, 15L/33R or H in use as appropriate.

In IMCs or when conditions are such that it does not appear likely that the pilot will complete the flight in accordance with the paragraph above:

DEPARTURE AIRCRAFT

- 1) Squawk 7600.
- 2) **MAINTAIN** the last assigned speed and level, or minimum flight altitude if higher, for a period of 7 minutes following:
 - i) The time the transponder is set to Code 7600; or
 - ii) The time the last assigned level or minimum flight altitude is reached; whichever is later and thereafter adjust level and speed in accordance with the filed flight plan;
- 3) When being vectored or having been directed by ATC, proceed in the most direct manner possible to rejoin the current flight plan route no later than the next significant point, taking into consideration the applicable minimum flight altitude.

RADAR 3A

No radio procedure: If no radio contact with ATC, squawk 7600, and unless otherwise instructed by ATC proceed as follows.

- 1) Fly SOUTH to join SEL R-269 to SEL VOR or proceed direct to YD100 then SEL VOR for G-597.
- 2) Fly SOUTH to join G-597 to NOPIK.
- 3) Fly SOUTH to YD100, YD130 and OSPOT for A-582.
- 4) Fly SOUTH to YD100, YD130, YD160, YD190 and BOPTA for BEDES via Z-51.

Pilot shall remain within D16.0 NCN (D40 SEL or 17 NM from airport as appropriate) while on heading 270°.

RADAR 3D

No radio procedure: If no radio contact with ATC, squawk 7600, and unless otherwise instructed by ATC proceed as follows.

- 1) Fly SOUTH to join SEL R-269 to SEL VOR or proceed direct to YD100 then SEL VOR for G-597.
- 2) Fly SOUTH to join G-597 to NOPIK.
- 3) Fly SOUTH to YD100, YD130 and OSPOT for A-582.
- 4) Fly SOUTH to YD100, YD130, YD160, YD190 and BOPTA for BEDES via Z-51.

Pilot shall remain within D16.0 NCN (D40 SEL or 17 NM from airport as appropriate) while on heading 250°.

RADAR 5S

No radio procedure: If no radio contact with ATC, squawk 7600, and unless otherwise instructed by ATC proceed as follows.

- 1) To SOT VOR via SOT R-315 for A-582, B-576.
- 2) To intercept Y-644 to POLEG, EGOBA, KARBU for G-597
- 3) To intercept SOT R-290 to BINIL or proceed direct to HP100, HP120, BELTU to BINIL for G-597 as appropriate.

Pilot shall proceed as described above prior to reaching D20 NCN or SEL R-235 or D16 WNG or D19 SOT due to airspace.

RADAR 3U

No radio procedure: If no radio contact with ATC, squawk 7600, and unless otherwise instructed by ATC proceed as follows.

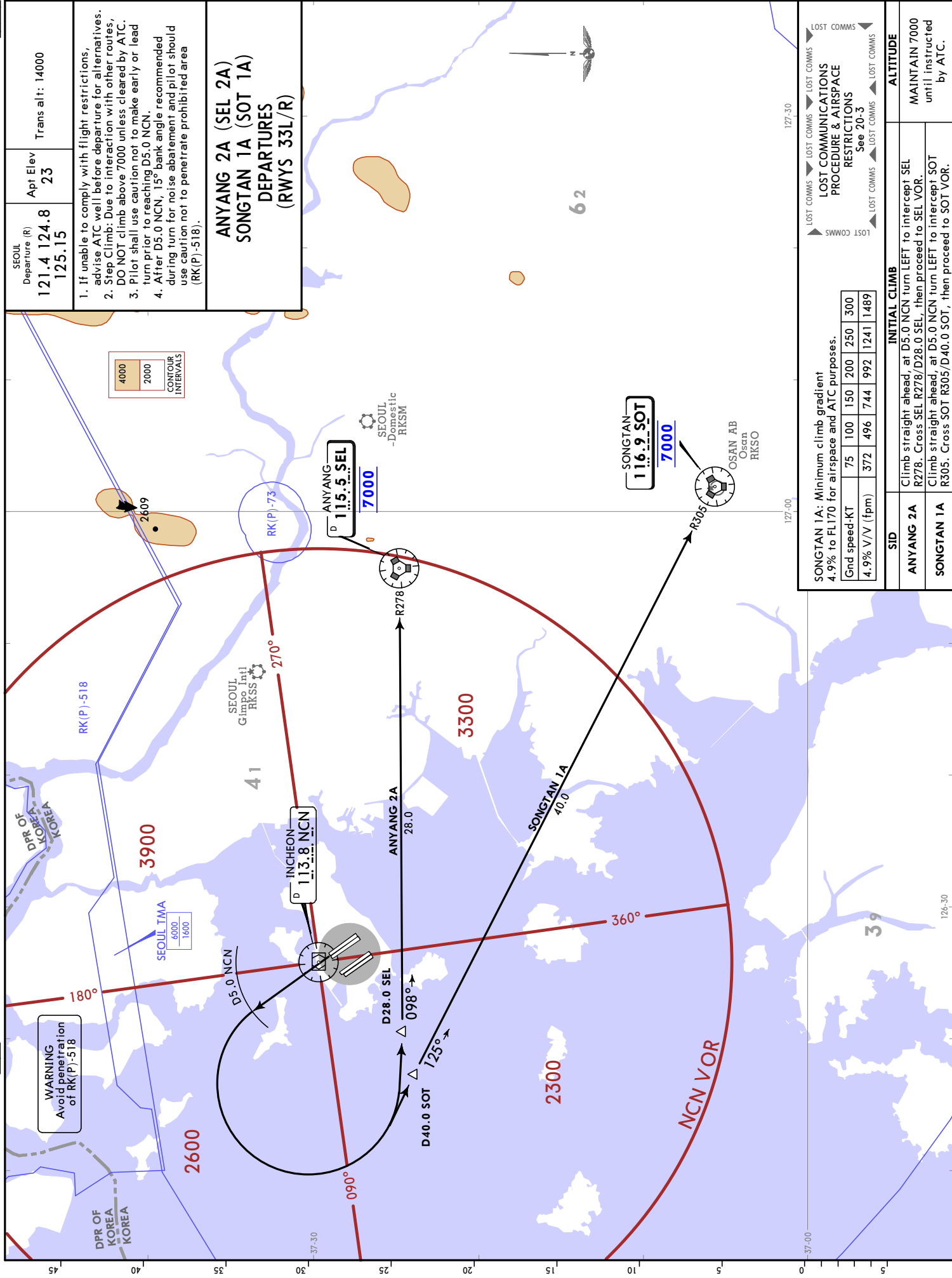
- 1) To SOT VOR via SOT R-307 for A-582, B-576.
- 2) To intercept Y-644 to POLEG, EGOBA, KARBU for G-597.
- 3) To intercept SOT R-290 to BINIL or proceed direct to HP100, HP120, BELTU to BINIL for G-597 as appropriate.

Pilot shall proceed as described above prior to reaching D14 WNG or SEL R-250 or D17 NCN or SOT R-300 due to airspace.

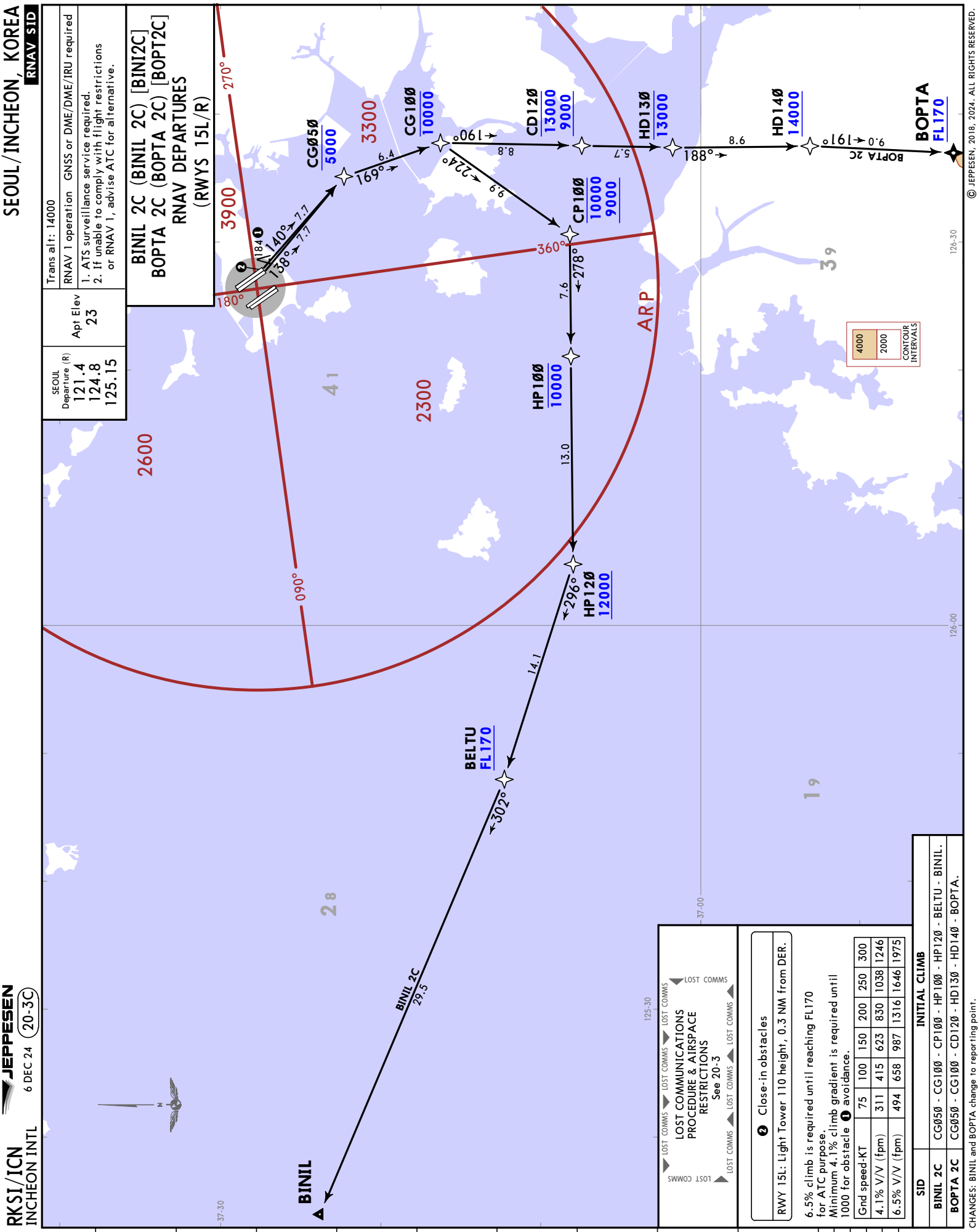
AIRSPACE RESTRICTIONS

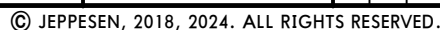
WARNING RK(P)-73

If an aircraft is seen flying towards RK(P)-73 without proper clearance, a tracer warning shot will be fired. If the aircraft continues into RK(P)-73 it will be shot down without further warning. An exception to this will be civilian aircraft which has been identified as friendly.









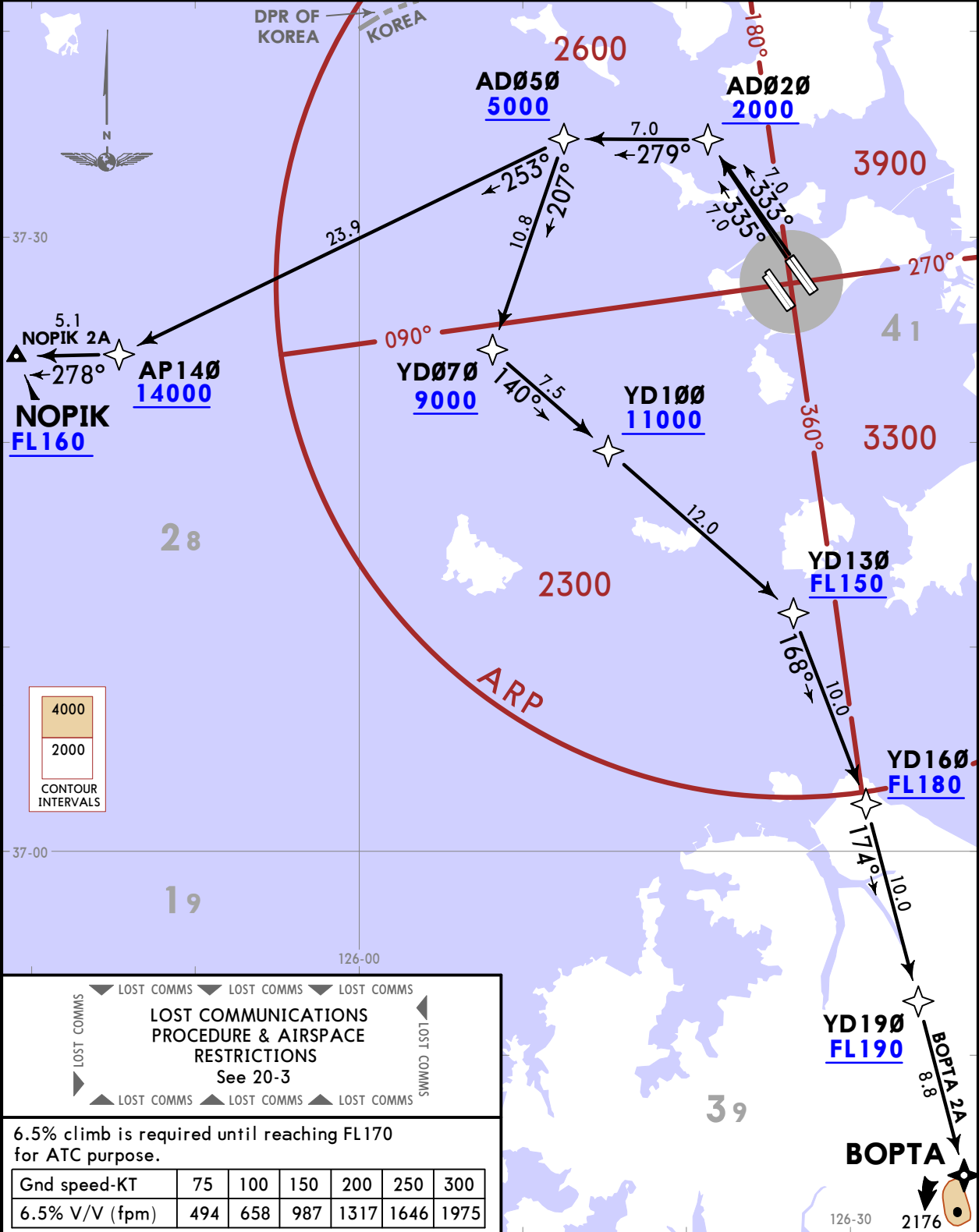
RKSI/ICN
INCHEON INTL

6 DEC 24 (20-3E)

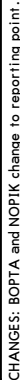
SEOUL/INCHEON, KOREA
RNAV SID

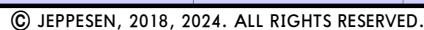
SEOUL Departure (R) 121.4 124.8 125.15	Apt Elev 23	Trans alt: 14000
		RNAV 1 operation GNSS or DME/DME/IRU required
		1. ATS surveillance service required. 2. If unable to comply with flight restrictions or RNAV 1, advise ATC for alternative.

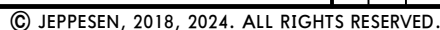
BOPTA 2A (BOPTA 2A) [BOPT2A]
NOPIK 2A (NOPIK 2A) [NOPI2A]
RNAV DEPARTURES
(RWYS 33L/R)



SID	INITIAL CLIMB
BOPTA 2A	AD020 - AD050 - YD070 - YD100 - YD130 - YD160 - YD190 - BOPTA.
NOPIK 2A	AD020 - AD050 - AP140 - NOPIK.





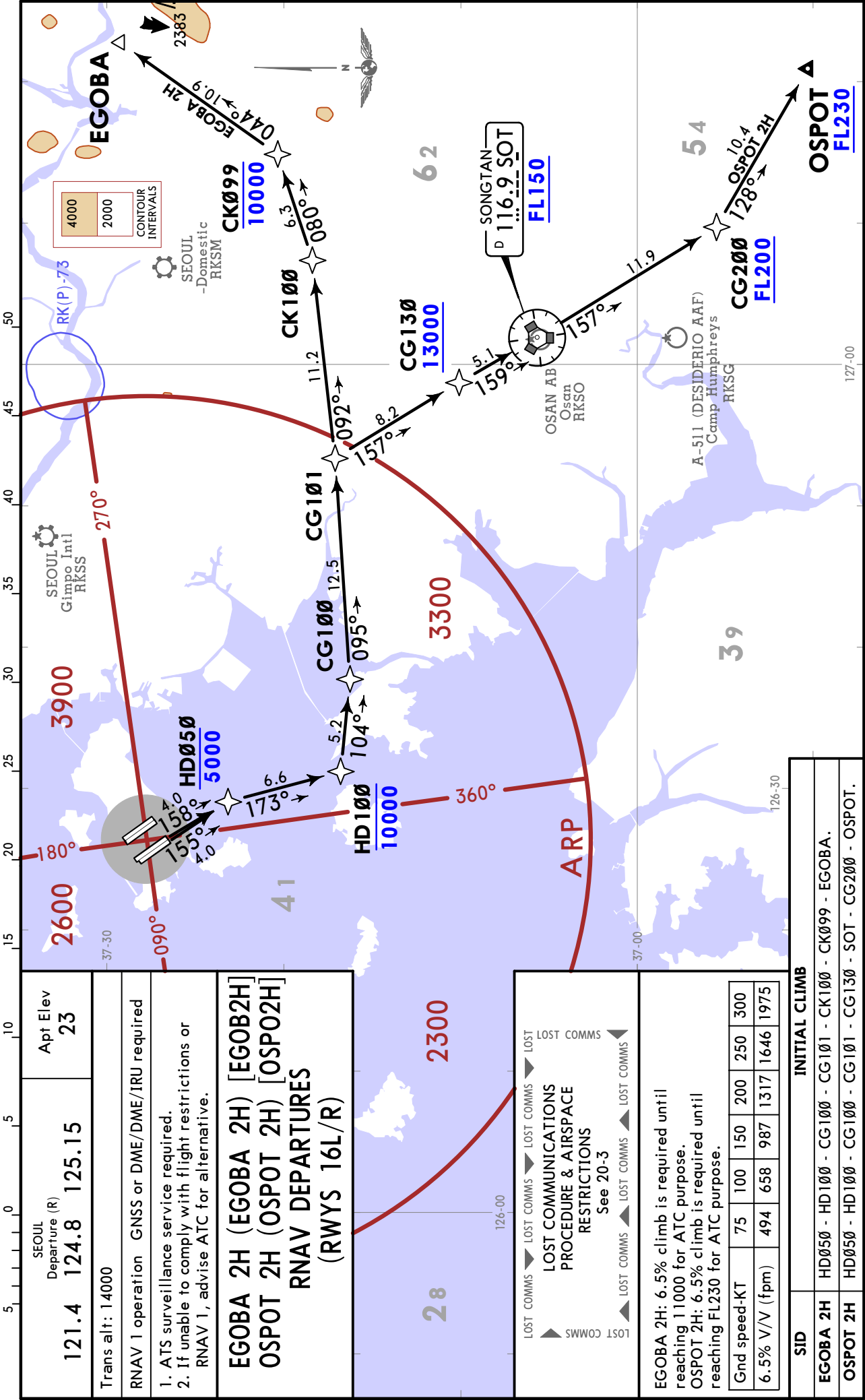


RKSI/ICN
INCHEON INTL

6 DEC 24 (20-3K)

SEOUL/INCHEON, KOREA

RNAV SID







RKSI/ICN
INCHEON INTL

 **JEPPESSEN**
2 AUG 24
Eff 7 Aug 1600Z (20-4A)

SEOUL/INCHEON, KOREA
NOISE

NOISE ABATEMENT

Local Time minus 9 HOURS = UTC

AIRCRAFT OPERATING PROCEDURES
(except helicopters)

TAKE-OFF

All departing aircraft should apply ICAO PANS-OPS (Doc 8168) Volume III Noise Abatement Take-off Climb Procedures as follows:

- 1. Runway 33 L/R, 34L/R:
 - Noise Abatement Departure Procedure ONE (NADP ONE)
 - a. Thrust reduction at 1500 ft above aerodrome elevation recommended.
- 2. Runway 15 L/R, 16L/R:
 - Noise Abatement Departure Procedure ONE or TWO (NADP ONE or NADP TWO)
 - a. NADP ONE: Thrust reduction at 1500 ft above aerodrome elevation recommended.
 - b. NADP TWO: Acceleration at 1000 ft above aerodrome elevation recommended.
 - c. For noise abatement and CO2 reduction using a NADP TWO is recommended.
 - If for safety reasons (prevention of bird strike), compliance with the recommended procedure is not possible, NADP ONE may be used.

AUXILIARY POWER UNITS (APUs)

- 1. At Passenger docking stands:
 - Primarily the stationary airport pneumatic and electrical service units shall be used.
 - Alternatively the airport owned mobile units shall be used.
- 2. At other stands:
 - The airport owned mobile units shall be used.
- 3. Airborne APUs shall only be started:
 - To start engine, the earliest 30 minutes before off-block time; however wide fuselage aircraft are permitted to use APU 60 minutes prior to scheduled departure time.
 - If maintenance work on the aircraft makes it unavoidable; in that case the service period shall be kept as short as possible.
 - If the airport owned units are not available or unserviceable for specific aircraft types; in that case the airborne APUs shall be started at the earliest 60 minutes before off-block time and be kept in operation not more than 30 minutes after the on-block time.

Note: In particular cases the Airport Corporation may permit longer service periods for APUs after the on-block time.

- Airport Corporation Telephone: 032-741-2458/9.
- INCHEON APRON CONTROL: 121.65 Mhz, 122.175 Mhz, 121.8 Mhz, 123.325 Mhz, 123.675 Mhz, 129.725 Mhz

RKSI/ICN

JEPPESEN

9 MAY 25

Eff 14 May 1600Z

(20-8A)

SEOUL/INCHEON, KOREA
INCHEON INTL

RWY 15L/33R AND TWYS CLOSED
DUE TO CONSTRUCTION
(SUP 039/25 AIRAC)

1. INTRODUCTION

- (1) Rwy 15L/33R will be closed due to pavement construction.
- (2) Some parts of Twy D will be closed due to pavement construction.

2. CLOSED AREAS & PERIOD (Refer to Rwy 15L/33R and Twys closed charts series)

Closed Areas		Period (UTC)
A	Rwy 15L/33R between Twy J and L * No effect on Twy J and Twy L operation.	From 1600 UTC 20 MAR 2025 To 2200 UTC 30 JUN 2025
	Rapid exit Twy C1, C2, C4, C5	
	Rapid exit Twy D1, D6	
	Twy K between Rwy 15R/33L and Twy D * No effect on Rwy 15R/33L operation.	
	Twy C between Twy K and L * No effect on Twy L operation.	
	Twy D between Twy D4 and L * No effect on Twy D4 and Twy L operation.	

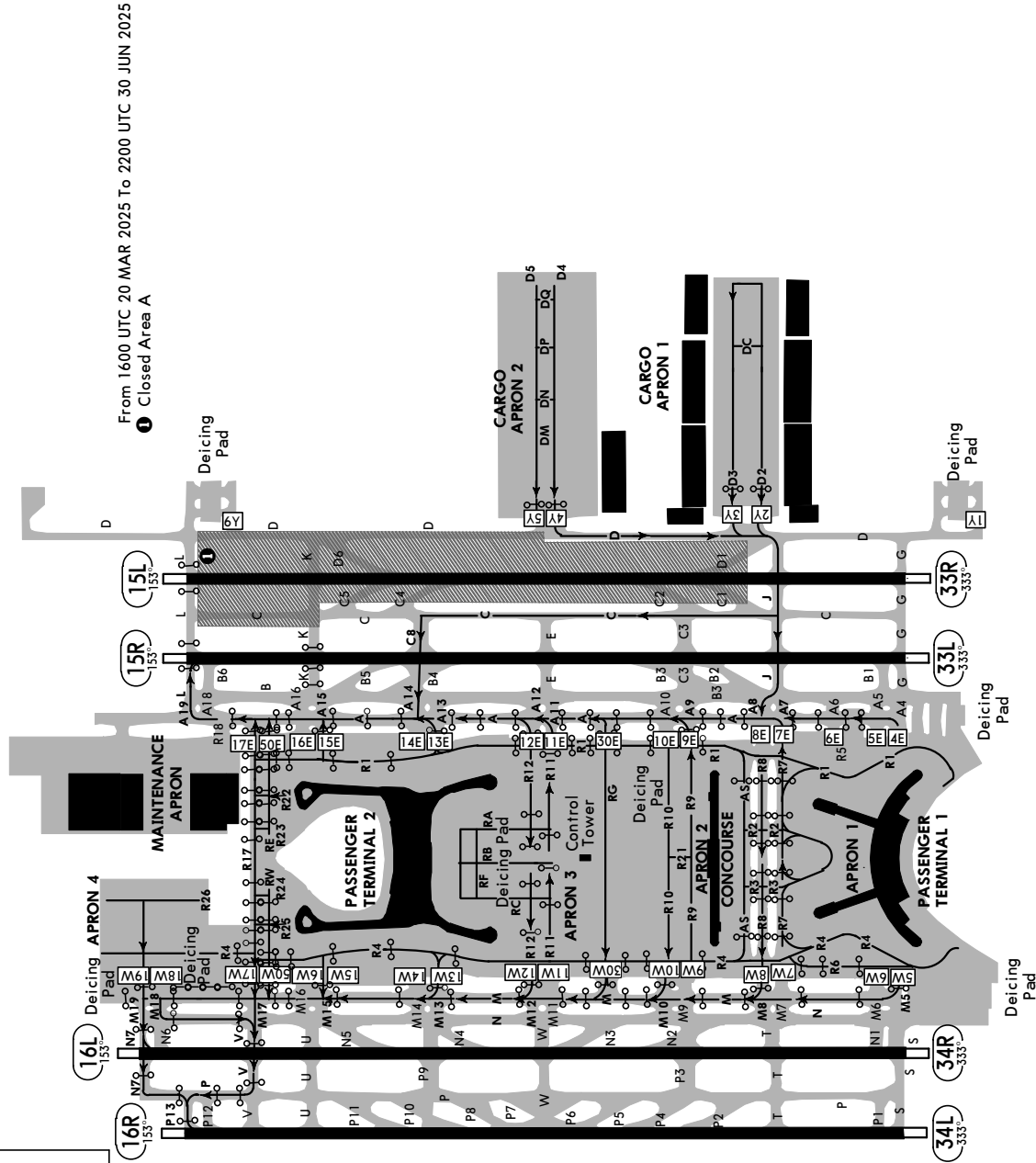
3. REMARKS

- (1) Rwy 15L/33R and Twys will be closed as shown in the charts.
- (2) Unserviceability lights and frangible barriers will be installed on the closed area.
- (3) All lights including runway centerline lights, runway edge lights, runway touchdown lights, approach lighting system for Rwy 15L/33R will be unserviceable.
- (4) Some parts of standard taxi routes (arrival and departure taxi routes charts) and low visibility procedures (low visibility taxi routes charts) will be temporarily changed due to the construction.
- (5) Temporary taxi routes may be changed by the ATC instruction.
- (6) Twy centerline lights, Twy edge lights, stop bar lights and taxiway guidance sign for the closed area will not be operated.
- (7) Works by equipment and vehicles will be conducted in the working area as shown in the charts.
- (8) Closed runway lighting is operated from 0800 to 2300 UTC at the thresholds of Rwy 15L/33R.
- (9) Any change to the contents of these pages will be notified by NOTAM.

TEMPORARY TAXI ROUTES

Departure for Rwy 15L/R, 16L/R and SMGCS - Departure for Rwy 15L/R, 16L/R

Aircraft Low Visibility Procedure Routes in Apron area can be used in both directions under the traffic situation. Taxiing Routes for A380 Aircraft are restricted in some Apron area. Refer to chart SPCL NOTICE TO A380 & B747-8 OPRS (CONTD 1). Taxiway RE&RW can only be used by Code C aircraft or smaller (maximum wingspan 118' (36m)).



LEGEND

CLOSED AREA

STOP BAR LIGHT

TAXIWAY/TAXI ROUTES DESIGNATION

TRANSFER OF CONTROL POINT

CHANGES: Applicable period.

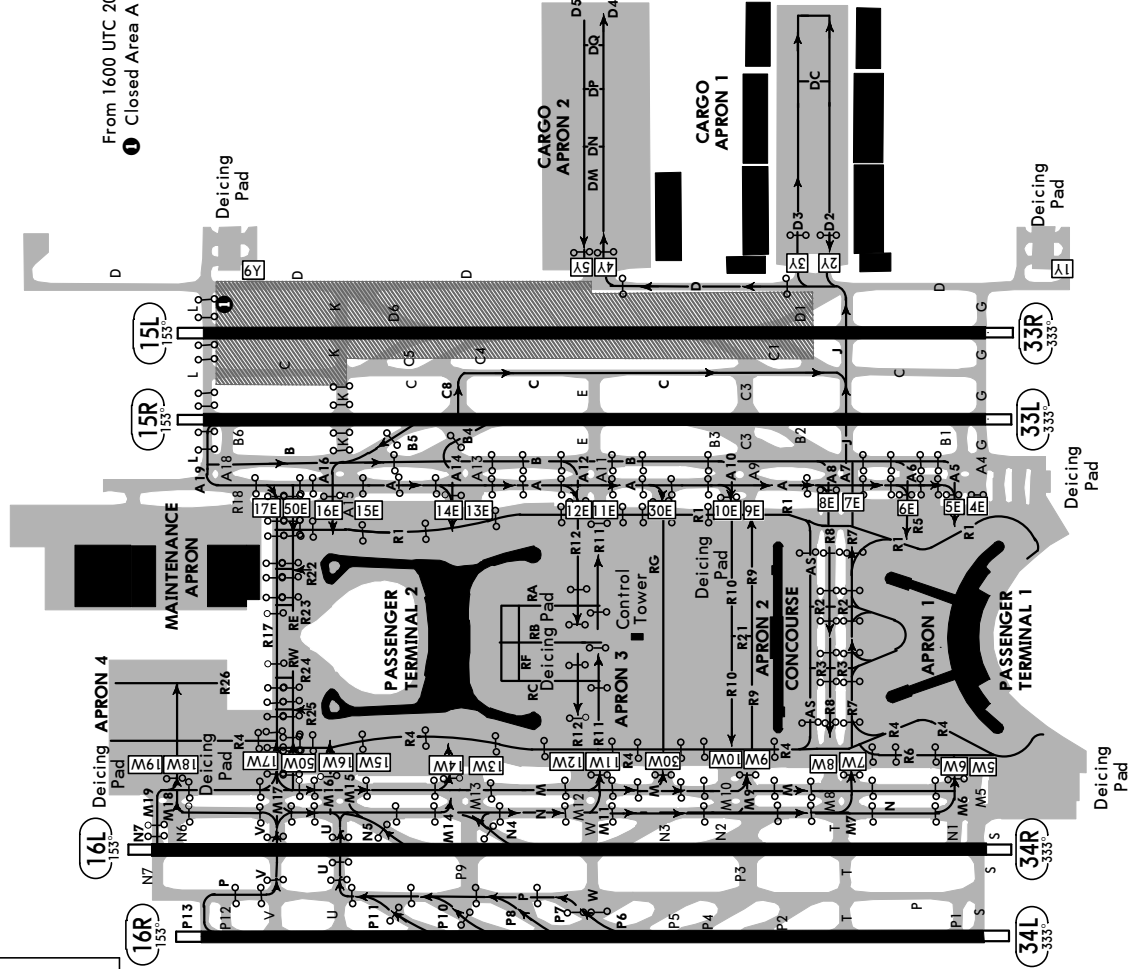
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TEMPORARY TAXI ROUTES

Arrival for Rwy 33L/R, 34L/R and SMGCS - Arrival for Rwy 33L/R, 34L/R

Aircraft Low Visibility Procedure Routes in Apron area can be used in both directions under the traffic situation. Taxiing Routes for A380 Aircraft are restricted in some Apron area. Refer to chart SPCL NOTICE TO A380 & B747-8 OPRS (CONTD 1). Taxiway RE&RW can only be used by Code C aircraft or smaller (maximum wingspan 118' (36m)).

From 1600 UTC 20 MAR 2025 To 2200 UTC 30 JUN 2025
● Closed Area A



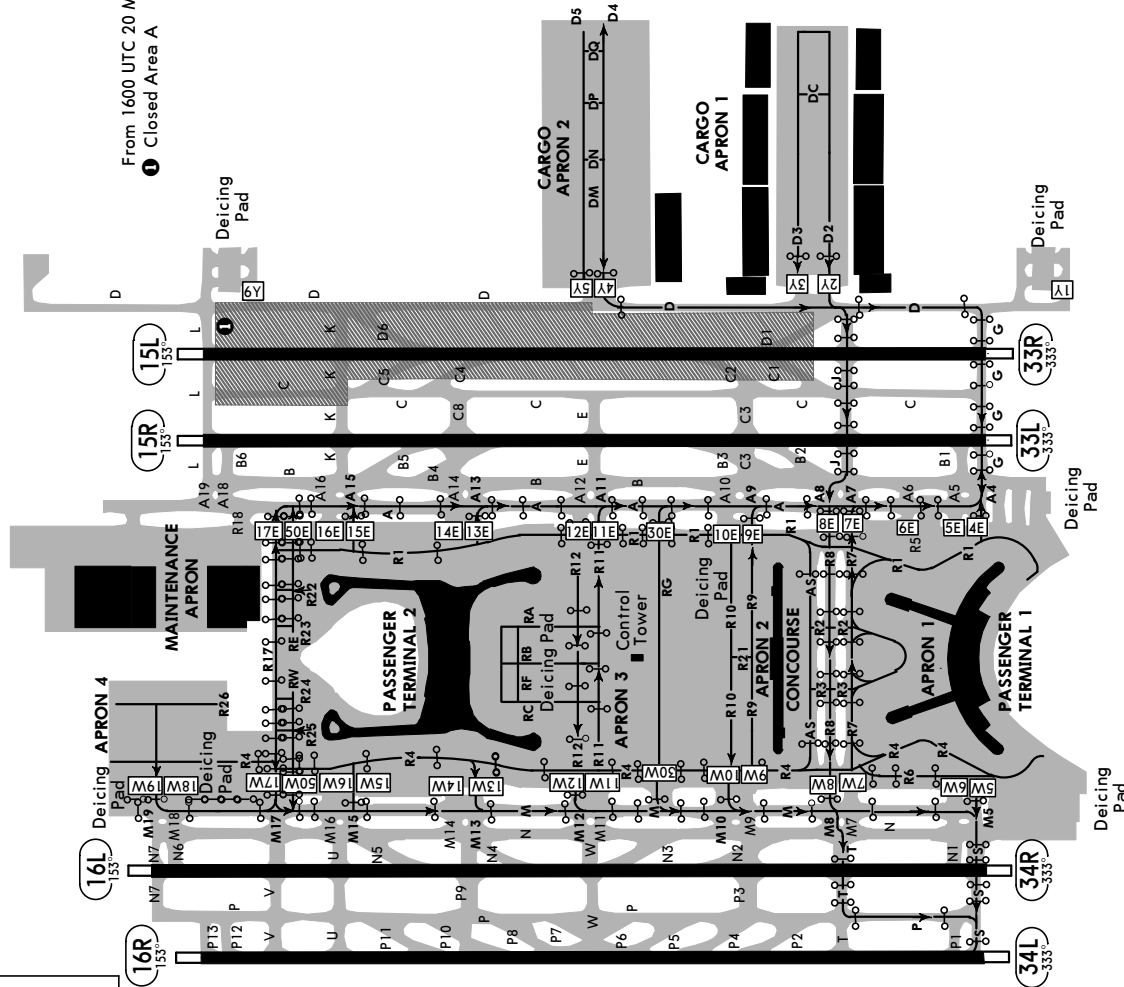
LEGEND

- CLOSED AREA
- TAXI ROUTE
- STOP BAR LIGHT
- TAXIWAY/ TAXI ROUTES DESIGNATION
- TRANSFER OF CONTROL POINT

TEMPORARY TAXI ROUTES

Departure for Rwy 33L/R, 34L/R and SMGCS - Departure for Rwy 33L/R, 34L/R

Aircraft Low Visibility Procedure Routes in Apron area can be used in both directions under the traffic situation. Taxiing Routes for A380 Aircraft are restricted in some Apron area. Refer to chart SPCL NOTICE TO A380 & B747-8 OPRS (CONTD 1). Taxiway RE&RW can only be used by Code C aircraft or smaller (maximum wingspan 118' (36m)).



LEGEND

- CLOSED AREA
- STOP BAR LIGHT
- TAXIWAY/TAXI ROUTES DESIGNATION
- TRANSFER OF CONTROL POINT

RKSI/ICN



SEOUL/INCHEON, KOREA
INCHEON INTL

INCHEON INTL AIRPORT UNDER MAINTENANCE OF
VDGS SERVICE FOR CONCOURSE
(SUP 003/25)

1. INTRODUCTION

Under maintenance of VDGS service for concourse.

2. VDGS CLOSED STANDS & PERIOD

No.	Parking Stand	VDGS Closed Period	Remarks
1	101 thru 115 (except 103 and 112)	From 0000 UTC 3 FEB 2025 To 1500 UTC 30 JUN 2025	A-CDM information to be provided From 1500 UTC 30 JUN 2025 To 1500 UTC 31 AUG 2025
2	117 thru 132 (except 122, 124 and 132)	From 0000 UTC 1 APR 2025 To 1500 UTC 31 AUG 2025	
3	103 and 112	From 0900 UTC 9 JAN 2025 To 1500 UTC 30 JUN 2025	A-CDM information to be provided From 1500 UTC 30 JUN 2025 To 1500 UTC 31 AUG 2025
4	122, 124 and 132	From 0900 UTC 9 JAN 2025 To 1500 UTC 31 AUG 2025	

3. REMARKS

1. There will not be provided any information from VDGS which is docking information (remaining distance, left and right deviation, etc.) A-CDM (TOBT, TSAT) information.
2. The aircraft shall be guided by marshaller.
3. Any change to the content of this chart will be notified by NOTAM.

RKSI/ICN

 **JEPPESSEN**
23 MAY 25 (20-8C)

SEOUL/INCHEON, KOREA
INCHEON INTL

**OPERATIONAL RESTRICTION OF CARGO APRON 1
(SUP 41/25)**

1. INTRODUCTION

Some part of taxilane D3, taxilane DD and aircraft stands 616, 627 thru 636 will be closed due to pavement construction as follows.

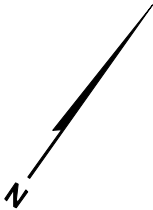
2. PERIOD AND WORKING AREAS

Closed Area			Period	Remarks (Aircraft Stand Availability)
Nr	Taxilane	Aircraft Stands		
❶	Some part of D3	627, 628, 629	From 2200 UTC 10 FEB 2025 To 0900 UTC 30 APR 2025	626 will be up to code D aircraft
❷		630, 631	From 2200 UTC 20 FEB 2025 To 0900 UTC 20 MAY 2025	632 will be up to code D aircraft
❸		632, 633, 634, 635	From 2200 UTC 30 APR 2025 To 0900 UTC 30 JUN 2025	631 will be up to code C aircraft
❹	DD	616, 636	From 2200 UTC 20 MAY 2025 To 1600 UTC 9 JUL 2025	-

3. REMARKS

- (1) Works by equipment and vehicles will be conducted in the area as shown in diagrams.
- (2) Safety fence will be installed to indicate the working area.
- (3) Equipments and vehicles do not penetrate obstacle limitation surfaces.
- (4) During working period, pilots should exercise caution and follow ATC's instruction when taxiing and push back near the working area.
- (5) Any change to the contents of this chart will be notified by NOTAM.

4. DIAGRAM



LEGEND

❶

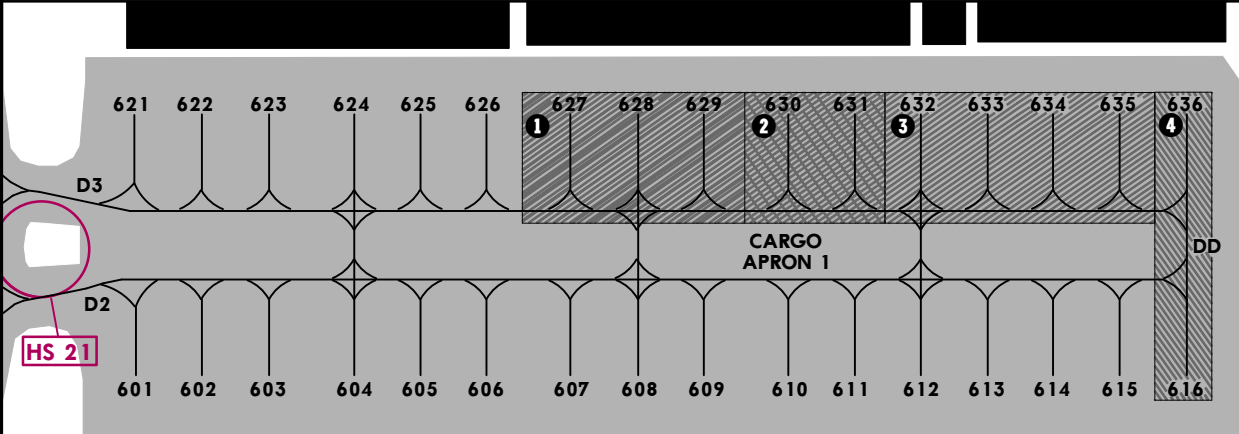
Closed Area

○

HS 21

Hot Spots

See AIRPORT INFO (CONTD),
TAKE-OFF MNMS
for description of Hot Spots



RKSI/ICN

 **JEPPESSEN**

6 JUN 25

Eff 11 Jun 1600Z

20-8D

SEOUL/INCHEON, KOREA
INCHEON INTL

**RWY 15L/33R AND TWYS CLOSED
DUE TO CONSTRUCTION
(SUP 044/25 AIRAC)**

1. INTRODUCTION

- (1) Rwy 15L/33R will be closed due to pavement construction.
- (2) Some parts of Twys C, D, L will be closed due to pavement construction.

2. CLOSED AREAS & PERIOD (Refer to Rwy 15L/33R and Twys closed D charts series)

Closed Areas		Period (UTC)
A	Rwy 15L/33R between Twy K and L * No effect on Twy K operation.	From 2200 UTC 30 JUN 2025 To 1600 UTC 09 JUL 2025
	Twy C between Twy K and L * No effect on Twy K operation.	
	Twy L between Rwy 15R/33L and Twy D * No effect on Rwy 15R/33L and Twy D operation.	
B	Rwy 15L/33R between Twy J and K * No effect on Twy J and Twy K operation.	
	Twy D between Twy D4 and K * No effect on Twy D4 and Twy K operation.	
	Rapid exit Twy C1, C2, C4, C5	
	Rapid exit Twy D1, D6	

3. REMARKS

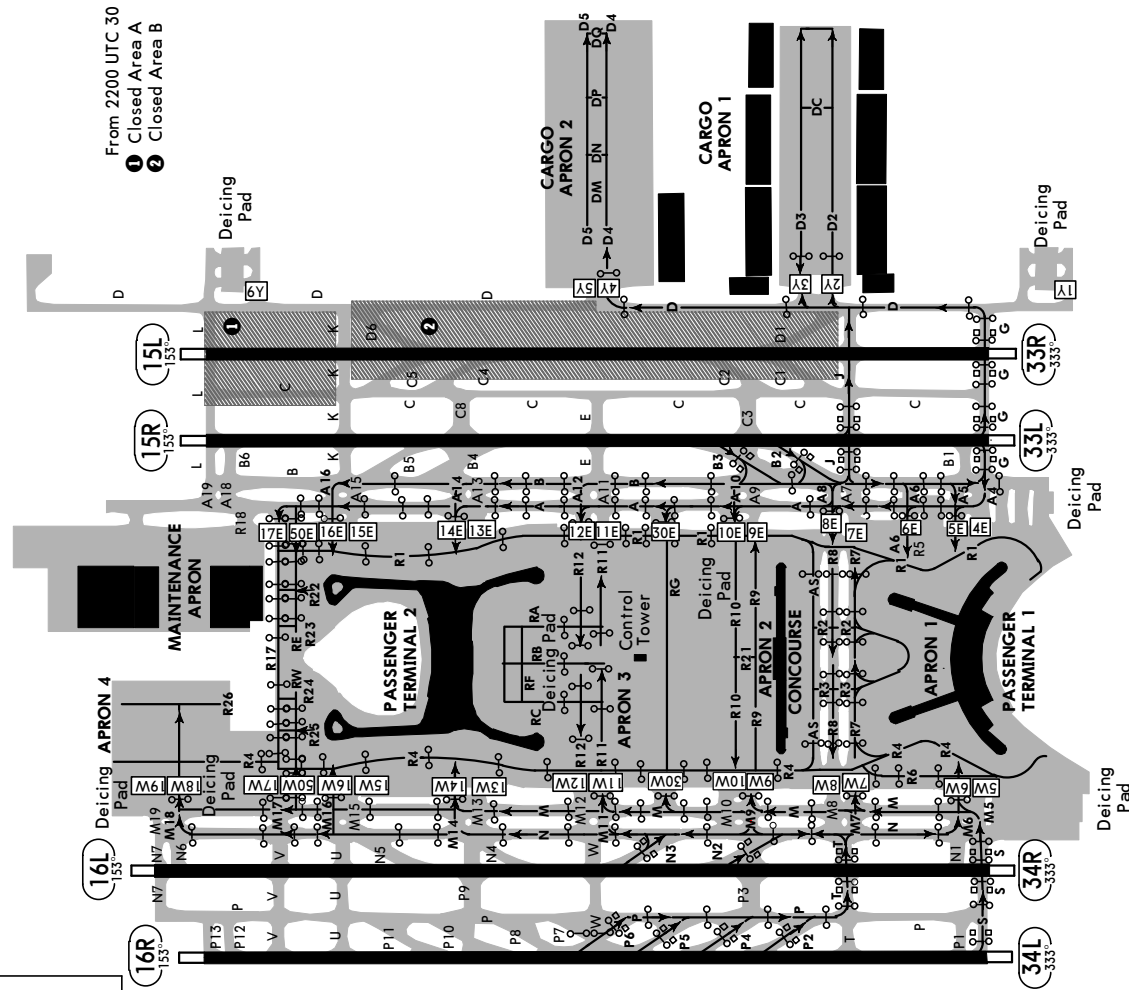
- (1) Rwy 15L/33R and Twys will be closed as shown in the Rwy 15L/33R and Twys closed D series charts.
- (2) Unserviceability lights and frangible barriers will be installed on the closed area.
- (3) All lights including runway centerline lights, runway edge lights, runway touchdown lights, approach lighting system for Rwy 15L/33R will be unserviceable.
- (4) Some parts of standard taxi routes (arrival and departure taxi routes charts) and low visibility procedures (low visibility taxi routes charts) will be temporarily changed due to the construction.
- (5) Temporary taxi routes may be changed by the ATC instruction.
- (6) Twy centerline lights, Twy edge lights, stop bar lights and taxiway guidance sign for the closed area will not be operated.
- (7) Works by equipment and vehicles will be conducted in the working area as shown in the charts.
- (8) Closed runway lighting is operated from 0800 to 2300 UTC at the thresholds of Rwy 15L/33R.
- (9) Any change to the contents of these pages will be notified by NOTAM.

TEMPORARY TAXI ROUTES

Arrival for Rwy 15L/R, 16L/R and SMGCS - Arrival for Rwy 15L/R, 16L/R

Aircraft Low Visibility Procedure Routes in Apron area can be used in both directions under the traffic situation. Taxiing Routes for A380 Aircraft are restricted in some Apron area. Refer to chart SPCL NOTICE TO A380 & B747-8 OPRS (CONTD 1). Taxiway RE&RW can only be used by Code C aircraft or smaller (maximum wingspan 118' (36m)).

From 2200 UTC 30 JUN 2025 To 1600 UTC 09 JUL 2025
1 Closed Area A
2 Closed Area B



LEGEND

CLOSED AREA

STOP BAR WITH RWY GUARD LIGHT

TAXIWAY// TAXI ROUTES DESIGNATION

STOP BAR

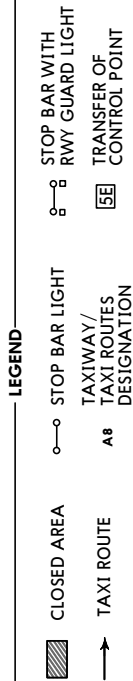
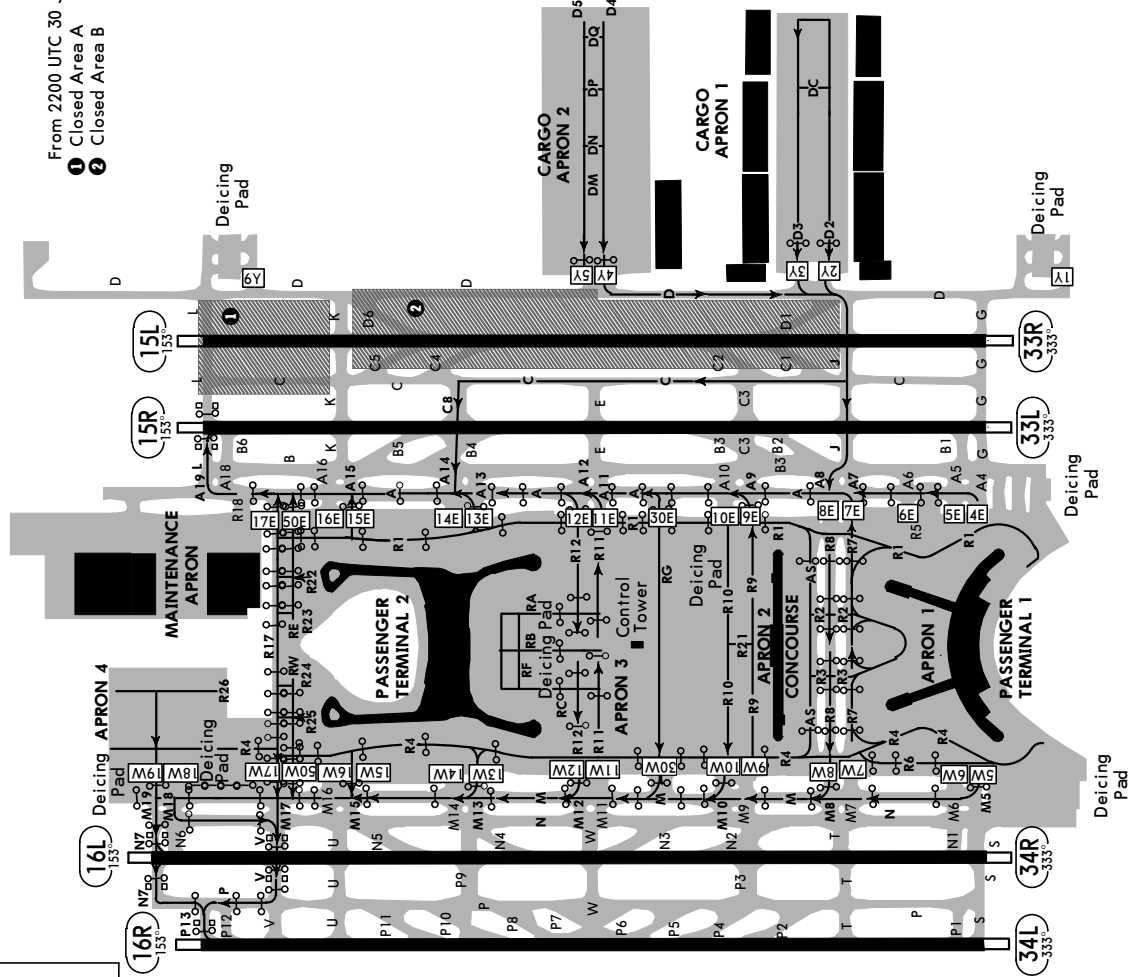
TRANSFER OF CONTROL POINT

TEMPORARY TAXI ROUTES

Departure for Rwy 15L/R, 16L/R and SMGCS - Departure for Rwy 15L/R, 16L/R

Aircraft Low Visibility Procedure Routes in Apron area can be used in both directions under the traffic situation. Taxiing Routes for A380 Aircraft are restricted in some Apron area. Refer to chart SPCL NOTICE TO A380 & B747-8 OPRS (CONTD 1). Taxiway RE&RW can only be used by Code C aircraft or smaller (maximum wingspan 118' (36m)).

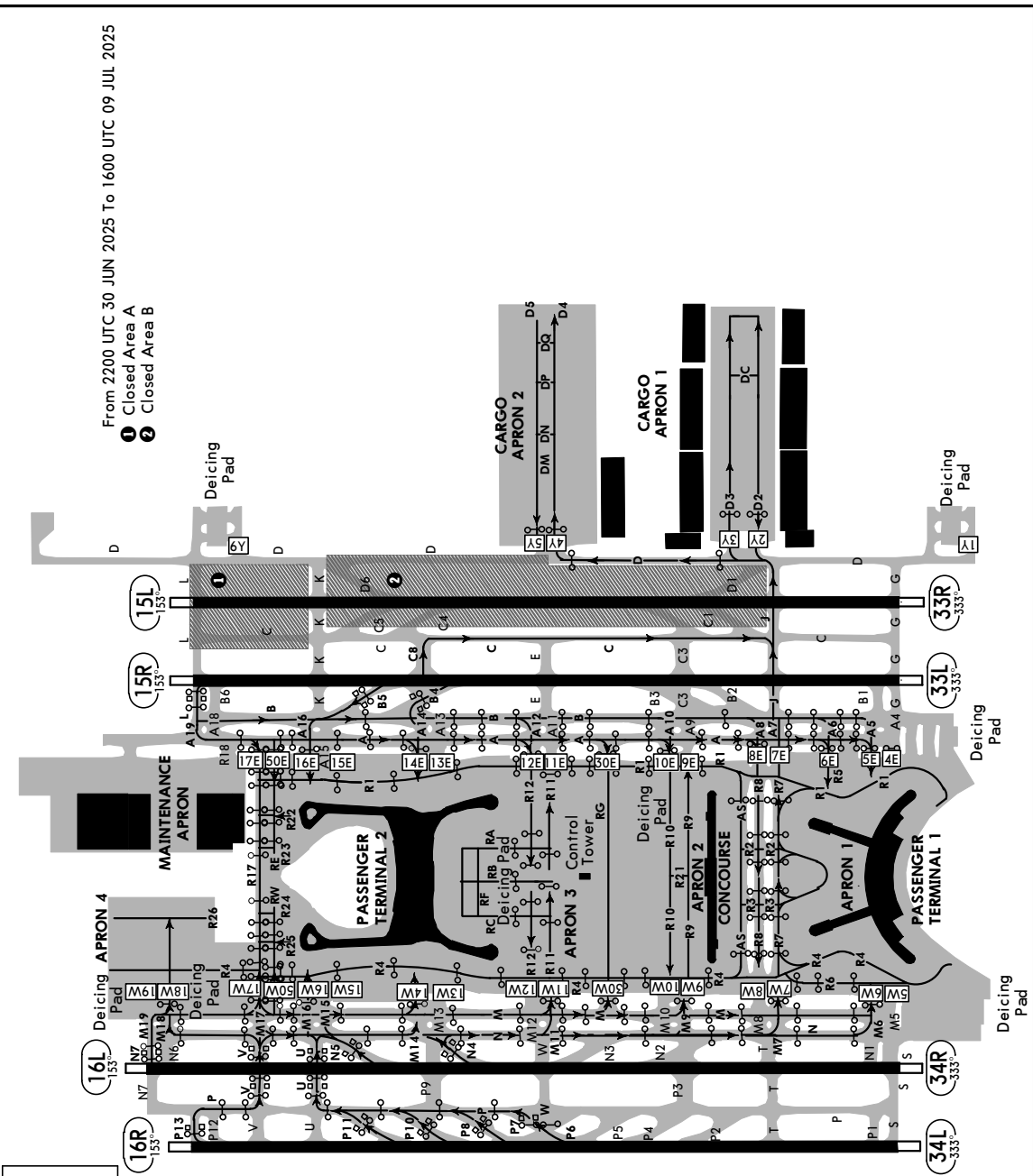
From 2200 UTC 30 JUN 2025 To 1600 UTC 09 JUL 2025
1 Closed Area A
2 Closed Area B



TEMPORARY TAXI ROUTES

Arrival for Rwy 33L/R, 34L/R and SMGCS - Arrival for Rwy 33L/R, 34L/R

Aircraft Low Visibility Procedure Routes in Apron area can be used in both directions under the traffic situation. Taxiing Routes for A380 Aircraft are restricted in some Apron area. Refer to chart SPCL NOTICE TO A380 & B747-8 OPRS (CONTD 1). Taxiway RE&RW can only be used by Code C aircraft or smaller (maximum wingspan 118' (36m)).



LEGEND

CLOSED AREA

TAXI ROUTE

STOP BAR LIGHT

STOP BAR WITH RWY GUARD LIGHT

TAXIWAY/ TAXI ROUTES DESIGNATION

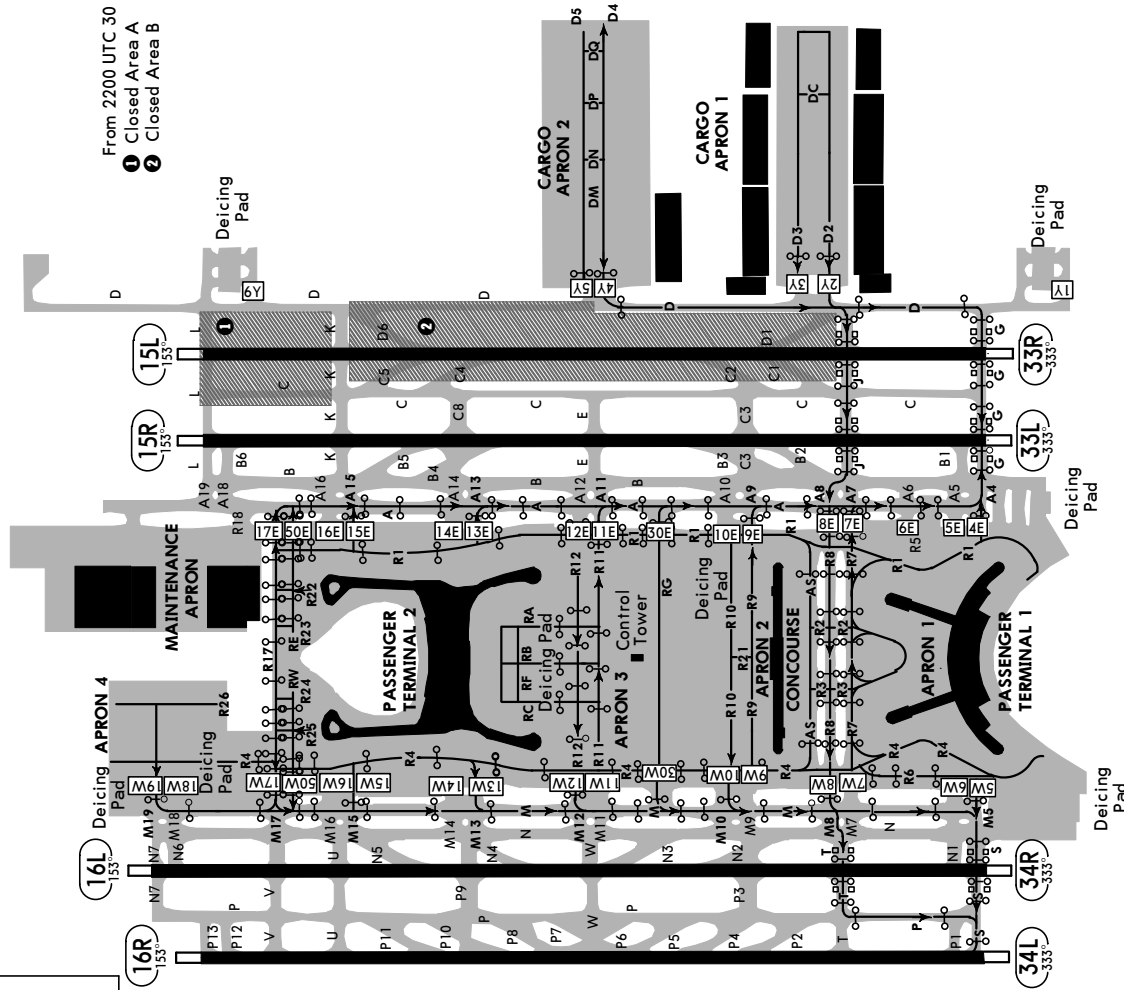
TRANSFER OF CONTROL POINT

TEMPORARY TAXI ROUTES

Departure for Rwy 33L/R, 34L/R and SMGCS - Departure for Rwy 33L/R, 34L/R

Aircraft Low Visibility Procedure Routes in Apron area can be used in both directions under the traffic situation. Taxiing Routes for A380 Aircraft are restricted in some Apron area. Refer to chart SPCL NOTICE TO A380 & B747-8 OPRS (CONTD 1). Taxiway RE&RW can only be used by Code C aircraft or smaller (maximum wingspan 118' (36m)).

From 2200 UTC 30 JUN 2025 To 1600 UTC 09 JUL 2025
1 Closed Area A
2 Closed Area B



LEGEND

	CLOSED AREA		STOP BAR WITH RWY GUARD LIGHT
	TAXI ROUTE		TAXIWAY/TAXI ROUTES DESIGNATION
	TRANSFER OF CONTROL POINT		

RKSI/ICN



SEOUL/INCHEON, KOREA
INCHEON INTL

**TRIAL OPERATION OF RE-CATEGORIZATION (RECAT) WAKE
TURBULENCE SEPARATION MINIMA WITHIN SEOUL TMA
(SUP 073/24)**

1. INTRODUCTION

As ICAO PANS-ATM revised in November 2020, RECAT wake turbulence separation minima will be used at Seoul TMA (RKSI, RKSS). The RECAT wake turbulence separation minima are based on a grouping of aircraft types into seven groups (A to G).

2. PERIOD

From 1600 UTC 14 DEC 2024 To 1600 UTC 15 DEC 2025.

3. APPLICATION

- (1) Applicable airport : Incheon INTL Airport (RKSI) and Gimpo INTL Airport (RKSS)
- (2) Applicable area : Seoul TMA, Incheon control zone, and Gimpo control zone
- (3) RECAT wake turbulence separation minima are applied for arrival and departure phases, when the aircraft is airborne.
- (4) For take-off phase, time based RECAT wake turbulence separation minima described in ICAO PANS-ATM is not applicable.
- (5) RECAT wake turbulence group designators are described as follows:

RECAT Group	MTOW	Wing Span	Example
GROUP A	299,829lbs (136,000kg) or more	245'(74.68m) < A < = 262'(80.0m)	A380
GROUP B		175'(53.34m) < B < = 245'(74.68m)	A359, B748, B773
GROUP C		125'(38.1m) < C < = 175'(53.34m)	B767, MD11
GROUP D	less than 299,829lbs (136,000kg), but more than 41,006lbs (18,600kg)	105'(32.0m) < D	A320, B737
GROUP E		90'(27.43m) < E < = 105'(32.0m)	E190, GLEX
GROUP F		F < = 90'(27.43m)	GLF4, CRJ7
GROUP G	41,006lbs (18,600kg) or less	No wing span criterion	C525, C172

- (6) The following separation minima will be provided between the leading aircraft and the succeeding aircraft as specified in 3.(6).a, 3.(6).b or 3.(6).c.

Leading Aircraft Group	Succeeding Aircraft Group	Separation Minima
A	B	4 NM
	C	5 NM
	D	5 NM
	E	6 NM
	F	6 NM
	G	8 NM
B	B	3 NM
	C	4 NM
	D	4 NM
	E	5 NM
	F	5 NM
	G	7 NM
C	D	3 NM
	E	3.5 NM
	F	3.5 NM
	G	6 NM
D	G	4 NM
E	G	4 NM

- a. An aircraft is operating directly behind another aircraft at the same altitude or less than 1000'(305m) below.
- b. Both aircraft are using the same runway, or parallel runways separated by less than 2493' (760m).
- c. An aircraft is crossing behind another aircraft, at the same altitude or less than 1000'(305m) below.

RKSI/ICN



SEOUL/INCHEON, KOREA

INCHEON INTL

**TRIAL OPERATION OF RE-CATEGORIZATION (RECAT) WAKE
TURBULENCE SEPARATION MINIMA WITHIN SEOUL TMA (CONTD)**

4. PILOT PROCEDURES

The application of RECAT wake turbulence separation minima will not affect the pilot procedure mostly. Nothing has changed with regard to flight planning and flight management. As the separation minima will be reduced in most cases, pilots should pay attention to the following points;

- (1) It is important to comply with ATC speed restriction at all times especially on the final approach course. The pilot must inform ATC as soon as possible if the speed cannot be maintained.
- (2) Pilots should report the type of the aircraft (including series) on the initial contact with Seoul Approach.
- (3) It is expected for pilots to vacate the runway expeditiously until the aircraft is completely clear of the runway.
- (4) The wake turbulence designator of the ICAO flight plan does not change. Pilots will continue to fill in the flight plan wake turbulence in item 9 with the ICAO aircraft category, H, M or L and J for SUPER category.
- (5) Pilots should include the word 'SUPER' or 'HEAVY' regarding the type of aircraft ICAO wake turbulence categories on the initial call. The suffix of 'SUPER' or 'HEAVY' has not been changed.
- (6) It is not required for pilots to know their RECAT wake turbulence aircraft group.

5. WAKE TURBULENCE ENCOUNTER REPORT

- (1) In order to conduct the safety assessment for wake turbulence separation minima by RECAT, it is required to collect Wake Turbulence Encounter Reports, and the information contained in the reports would be used only for the purpose of safety assessment.
- (2) Action to be taken by pilot:
When a pilot has encountered wake turbulence, the pilot should submit the Wake Turbulence Encounter Report.

6. REMARK

Any change to the contents of these charts will be notified by NOTAM.

RKSI/ICN

 **JEPPESSEN**
25 APR 25 (20-8G)

SEOUL/INCHEON, KOREA
INCHEON INTL

INCHEON INTERNATIONAL AIRPORT A-CDM TRIAL OPERATION FOR PHASE 2 (SUP 33/25)

1. INTRODUCTION

- (1) A-CDM is a process that allows air traffic controllers, airport operators, aircraft operators (AO), ground handling agents (GHA), pilots and air traffic flow managers to exchange operational information and work together to efficiently manage operations at aerodrome. A-CDM involves sharing of accurate and timely information amongst airport partners through airport systems and implementing a set of operational procedures and automated processes.
- (2) During A-CDM phase 1 trial operation, flights subject to A-CDM pre-departure sequencing were limited to ATFM regulated flights due to limited pre-departure sequencer (departure manager) function.
- (3) Pre-departure sequencer (departure manager) has been upgraded to implement pre-departure sequence to all departure flights, and hence A-CDM phase 2 trial operation will be implemented based on Incheon International A-CDM implementation plan.
- (4) The phase 2 trial operation will be introduced in 3 progressive steps to allow all airport partners to be involved an opportunity to review and refine their workflow process and relevant hardware system, especially pre-departure sequencer, prior to actual implementation.
- (5) The purpose of these charts is to provide information on the Incheon A-CDM system and operational procedures, and to detail the timings of the planned Incheon A-CDM trial operation for phase 2 and the required actions by pilots, aircraft operators (AO) and ground handling agents (GHA).

2. A-CDM PHASE 2 TRIAL OPERATION PROGRESSIVE STEPS

- (1) STEP 1 - From 1600 UTC 29 JUN 2022 To 1600 UTC 29 JUN 2023
 - a. Aircraft operators (AO) and ground handling agents (GHA) are required to access Incheon A-CDM system and input manual TOBT based on A-CDM pre-departure procedures paragraphs 3.(1) to 3.(2) below.
- (2) STEP 2 - From 1600 UTC 29 JUN 2022 To 1700 UTC 25 JUN 2025
 - a. Aircraft operators (AO) and ground handling agents (GHA) are required to access Incheon A-CDM system and input manual TOBT based on A-CDM pre-departure procedures paragraphs 3.(1) to 3.(2) below.
 - b. Pre-departure sequencer will be installed in ATC system and Site-Acceptance-Test will be done.
 - c. ATC will validate TTOT/TSAT calculation accuracy and check ATC HMI function using actual TOBT.
 - d. ATC can issue TSAT to departure flights only in a specific time zone based on A-CDM pre-departure procedures paragraphs below.
 - e. If ATC issue TSAT to all departure flights, the specific time zone will be noticed to aircraft operator (AO) and ground handling agents (GHA) through the following channels:
 - (a) A-CDM portal system; and
 - (b) Automatic Terminal Information Services (ATIS)
 - f. AO, GHA and pilot shall comply with TSAT and A-CDM pre-departure procedures paragraphs below.
- (3) STEP 3 - From 1700 UTC 25 JUN 2025 To 1600 UTC 23 JUL 2025
 - a. Aircraft operators (AO) and ground handling agents (GHA) are required to access Incheon A-CDM system and input manual TOBT based on A-CDM pre-departure procedures paragraphs below.
 - b. ATC will issue TSAT to all departure flights based on A-CDM pre-departure procedures paragraphs below.
 - c. AO, GHA and pilot shall comply with TSAT and A-CDM pre-departure procedures paragraphs below.
 - d. Airport performance indicator (TOBT accuracy, TSAT compliance and departure punctuality, etc.) will be monitored.

3. A-CDM PRE-DEPARTURE SEQUENCE PROCEDURES

- (1) Flight plan discrepancy check
 - a. In order to receive TSAT, flight data in Airport Operational Database (AODB) and ATC flight plan shall be identical. AO are required to manage flight data identical on those two systems.
Note - Call-sign, EOBT, DOF, ADES, ADEP items are used to match TOBT message with ATC flight plan in pre-departure sequencer.
- (2) Target Off-Block Time (TOBT)
 - a. Incheon airport A-CDM portal system will automatically calculate system TOBT for each departure flight taking into account the Estimated In-Block Time/Actual In-Block Time (EIBT/AIBT), Minimum Turnaround Time (MTTT) and Estimated Off Block Time (EOBT). AO or GHA may refer this system generated TOBT when input TOBT.
 - b. AO or GHA are required to confirm or update the system generated TOBT from 90 minutes to 40 minutes prior to TOBT. TOBT which is confirmed or updated will be applied to pre-departure sequencing to calculate TSAT.
 - c. If the prediction of departure readiness (new TOBT) differs more than 5 minutes (plus or minus) from the previous TOBT, AO or GHA shall update TOBT.

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INCHEON INTERNATIONAL AIRPORT A-CDM TRIAL OPERATION FOR PHASE 2 (CONTD)

(2) Target Off-Block Time (TOBT) (contd.)

- d. TOBT shall not deviate from EOBT by more than 15 minutes. If TOBT deviates from EOBT by more than 15 minutes, AO has to initiate an delay/change message. When EOBT is modified, TOBT is automatically modified to the value of EOBT. Even if when EOBT is modified, TOBT which is confirmed by AO or GHA will be remained.
- e. TOBT can be corrected as often as required up until the time the TSAT is issued (30 minutes prior to TOBT).
- f. The accuracy of TOBT is vital to an optimal TSAT. Thus AO or GHA are strongly encouraged to update TOBT as soon as any expected delay to the aircraft readiness for push-back is made available to avoid unnecessary hold-ups.
- g. After TSAT has been issued, TOBT may be corrected up to three times for stable TSAT operation. For the forth time update, TOBT has to be deleted and a new one has to be entered. In this case, TSAT may be delayed as TSAT slot is lost in pre-departure sequencing list.
- h. If AO wants to delay the passenger boarding start time due to gap between TOBT and TSAT caused by CTOT, traffic congestion, etc., TOBT has to be updated with the latest time.
- i. If it is impossible to take-off due to RVR minima or adverse weather condition, unable to predict the ground handling time, etc., TOBT must be deleted or updated with delayed time. And TOBT must be re-submitted whenever aircraft ready is predictable.
- j. TOBT shall be updated through the following channels:
 - (a) A-CDM portal and mobile APP; or
 - (b) Flight Information Assistant (FIA) at PBB boarding rooms; or
 - (c) A-CDM operation center
- k. TOBT information is available through the following channels:
 - (a) A-CDM portal and mobile APP; or
 - (b) Flight Information Assistant (FIA) at PBB boarding rooms; or
 - (c) Visual Docking Guidance System (VDGS); or
 - (d) Radio communication with AO or GHA
- l. AO or GHA are required to provide TOBT to pilot in case VDGS is unserviceable or in cargo apron/remote stand.

(3) Target Start-up Approval Time (TSAT) - without de-icing

- a. Pilot shall ensure aircraft is ready for push-back within 5 minutes of TOBT. If it is expected to differ by 5 minutes or more, pilot shall notify the AO or GHA to update TOBT.
- b. TSAT will be issued at TOBT -30 minutes.
- c. If the operation situation changes, Departure Manager (D-MAN) can update the TSAT already issued. AO or GHA has to monitor TSAT continuously before Actual Off-Block Time (AOBT).
- d. If TSAT is not issued at TOBT -30 minutes, AO or GHA has to take measures to make flight plan data of A-CDM portal system and ATC system identical.
- e. If TSAT cannot be complied with because the new TOBT is updated later than TSAT, new TSAT will be issued.
- f. If new TOBT is earlier than TSAT, TSAT may be improved only when free TSAT is available not affecting other flights TSAT.
- g. TSAT information is available through the following channels:
 - (a) A-CDM portal system; or
 - (b) Flight Information Assistant (FIA) at PBB boarding rooms; or
 - (c) Visual Docking Guidance System (VDGS); or
 - (d) Radio communication with GHA or AO; or
 - (e) Incheon Apron (in case VDGS is unserviceable.)
- h. AO or GHA are required to provide TSAT to pilot in case VDGS is unserviceable or in cargo apron/remote stand.

(4) Target Start-up Approval Time (TSAT) - regulated flight

- a. If a flight applied with CTOT is expected to be unable to comply with CTOT or TSAT due to AO or GHA internal issue, AO or GHA are required to consult with ATCC to get a new CTOT.
- b. If a flight applied with CTOT is expected to be unable to comply with CTOT or TSAT due to ATC issue, ATC will update TTOT/TSAT with new CTOT through consultation with ATCC.
- c. When a flight applied with CTOT cannot comply with TSAT but can comply CTOT, pilot shall notify ATC of this situation. In this case, the estimated push-back time may be notified specifically.

Note - When D-MAN receives CTOT, D-MAN calculate TSAT without considering TOBT.
For this reason, TSAT may be earlier than TOBT.
- d. When TSAT is issued by ATC not by D-MAN, TSAT may be earlier time than TOBT or may not be updated as TOBT updated. AO or pilot are required to notify ATC in the following cases:
 - (a) If TSAT cannot be complied with due to TSAT earlier than TOBT.
 - (b) If TSAT is not updated even though updating TOBT later than TSAT.

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 **JEPPESSEN**
28 FEB 25 (20-8G2)

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INCHEON INTERNATIONAL AIRPORT A-CDM TRIAL OPERATION FOR PHASE 2 (CONTD)

- (5) ATC clearance, start-up and push-back procedures
 - a. Pilot shall request ATC clearance from Incheon Delivery via RTF or DCL within TOBT -10 minutes to +5 minutes. If pilot do not request ATC clearance by TOBT +5 minutes, ATC will cancel TOBT and TSAT will be canceled. AO or GHA shall re-enter TOBT to receive TSAT.
 - b. Pilot shall request push-back from Incheon Apron within TSAT +/-5 minutes. If pilot do not request push-back by TSAT +5 minutes, ATC will cancel TOBT and TSAT will be cancelled. AO or GHA shall re-enter TOBT to receive TSAT.
 - c. Taxi clearance must be requested at roll-out positions by TSAT +10 minutes.
- (6) De-icing and A-CDM
 - a. Incheon INTL Airport starts to implement trial operation for the de-icing milestones in A-CDM program with D-MAN, indicating start/end times and duration of de-icing from winter 2024/2025.
 - b. If de-icing is required, AO or GHA has to request de-icing 40 minutes prior to TOBT through A-CDM portal system or mobile APP. As soon as the request for de-icing is received, Incheon De-icing will allocate remote de-icing zone. The pilot shall ensure aircraft is ready for push-back at TOBT. If pilot does not request push-back for de-icing within 5 minutes of TOBT, the order of pre-assigned de-icing can be changed.
 - c. Whenever a flight has been flagged for de-icing, TSAT will be generated based on Estimated de-icing Pad In Time(EPIT) taking into account the taxi time to the pad + a standard queueing time. During trial operation for the de-icing milestones (2024/2025), airport operator will validate TSAT for de-icing milestones accuracy so TSAT will be not shared to stake holders through channels (3.(3).g).
 - d. Estimated End of De-icing Time(EEZT) is a calculated element, derived from Estimated Commence of De-icing Time(ECZT) + Estimated De-Icing Time(EDIT). An update of EEZT is needed by GHA after the Actual de-icing Pad In Time(APIT) occurs with Integrated Information System(IIS) or mobile APP.
 - e. (Engine off) Once de-icing is completed, contact Incheon delivery to get ATC clearance. Report "Engine off de-icing and de-icing completed".
(Engine on) Once de-icing is started, contact Incheon delivery to get ATC clearance. Report "Engine on de-icing and de-icing started".
 - f. Cancellation of de-icing request is possible through the following channels:
 - (a) A-CDM portal and mobile app; or
 - (b) Communication with the pilot and Incheon De-icing.
 - g. For details on de-icing and de-icing procedures, refer to DEICING charts (20-9M1 thru 20-9M4).
- (7) Non A-CDM procedures
 - a. The non A-CDM procedures are applicable when A-CDM cannot be operated normally due to system issue or maintenance as follows:
 - (a) If AO or GHA is unable to submit TOBT in any channels;
 - (b) If Departure Manager (D-MAN) and Integrated Information System (IIS) cannot exchange TOBT and TSAT due to IIS system issues;
 - (c) If Departure Manager (D-MAN) has system issues;
 - (d) When switching from D-MAN main system to a back-up system.
 - b. Except paragraph 3.(6).a above, even if non A-CDM procedures are being applied, when it is available to input TOBT, AO or GHA are required enter TOBT continuously for prompt transition to normal operation.
 - c. When non A-CDM mode, the following procedures are applied for requesting ATC clearance and push-back.
 - (a) ATC clearance can be requested via voice RTF or Data-link Departure Clearance Service (DCL) from EOBT -10 minutes.
 - (b) The sequence of departure of take-off is determined by ATC.
 - (c) If a flight is unable to commence push-back within 10 minutes after receiving ATC clearance due to the aircraft being unready, ATC clearance and TSAT will be cancelled.

4. REMARKS

- (1) During trial operation for phase 2, TSAT will be provided to all departure flights only when Departure Manager (D-MAN) system operates normally.
- (2) Only TOBT which is confirmed or updated by AO or GHA will be applied to pre-departure sequencing for calculating TSAT.
- (3) Any change to the contents of these pages will be notified by NOTAM or AIP.